For the use only of a Registered Medical Practitioner

PANTOCID GR TABLETS 20 MG / 40

(Pantoprazole Sodium Gastro Resistant Tablets 20/40 mg)

COMPOSITION

PANTOCID GR TABLETS 20 MG

Each enteric-coated tablet contains:
Pantoprazole Sodium sesquihydrate
equivalent to Pantoprazole……………………20 mg

PANTOCID GR TABLETS 40 MG

Each enteric-coated tablet contains:
Pantoprazole Sodium sesquihydrate
equivalent to Pantoprazole……………………40 mg

Inactive ingredients: Anhydrous Sodium Carbonate, Mannitol, Crospovidone, Hydroxypropyl Cellulose – L, Purified Water, Microcrystalline Cellulose, Calcium Stearate, Opadry 02H52369 Yellow, Methacrylic Acid Ethyl Acrylate Copolymer (1:1) dispersion*, Triethyl Citrate, Sodium Lauryl Sulphate, Titanium Dioxide, Ferric Oxide Yellow, Purified Talc.

PRODUCT DESCRIPTION

PANTOCID GR TABLETS 20 MG

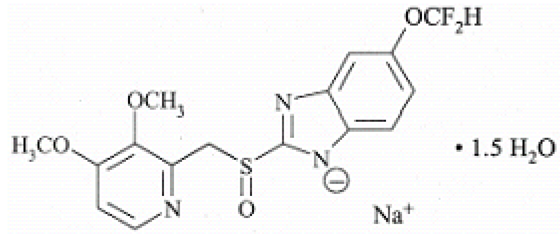
Yellow colored enteric-coated oval shaped, biconvex tablets plain on both sides. Size 8.8x4.5 mm.

PANTOCID GR TABLETS 40 MG

Yellow colored enteric-coated oval shaped, biconvex tablets plain on both sides. Size 11.7x5.9 mm

DESCRIPTION

PANTOCID GR TABLETS 20 / 40 MG contain pantoprazole sodium sesquihydtrate, which is a substituted benzimidazole. It inhibits gastric acid secretion. It is chemically designated as sodium 5-(difluoromethoxy)-2-[[(3, 4-dimethoxy-2-pyridinyl) methyl] sulfinyl]-1H-benzimidazole sesquihydrate. Its empirical formula is $C_{16}H_{14}F_2N_3NaO_4S$ x 1.5 H_2O, and its molecular weight is 432.4.



PANTOPRAZOLE

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- **Mechanism of Action**

Pantoprazole is a substituted benzimidazole which inhibits the secretion of hydrochloric acid in the stomach by specific action on the proton pumps of the parietal cells.

Pantoprazole is converted to its active form in the acidic canaliculi of the parietal cells where it inhibits the H^+, K^+-ATPase enzyme, i.e. the final stage in the production of hydrochloric acid in the stomach. The inhibition is dose-dependent and affects both basal and stimulated acid secretion. In most patients, freedom from symptoms is achieved within 2 weeks. As with other proton pump inhibitors and H_2 receptor inhibitors, treatment with pantoprazole causes a reduced acidity in the stomach and thereby an increase in gastrin in proportion to the reduction in acidity. The increase in gastrin is reversible. Since pantoprazole binds to the enzyme distal to the cell receptor level, the substance can affect hydrochloric acid secretion independently of stimulation by other substances (acetylcholine, histamine, gastrin). The effect is the same whether the product is given orally or intravenously.

The fasting gastrin values increase under pantoprazole. On short-term use, in most cases they do not exceed the normal upper limit. During long-term treatment, gastrin levels double in most cases. An excessive increase, however, occurs only in isolated cases. As a result, a mild to moderate increase in the number of specific endocrine (ECL) cells in the stomach is reported in a minority of cases during long-term treatment (simple to adenomatoid hyperplasia).~~,~~ However, carcinoid precursors (atypical hyperplasia) or gastric carcinoids have not been reported in humans as reported in animals.

An influence of a long term treatment with pantoprazole exceeding one year cannot be completely ruled out on endocrine parameters of the thyroid according to reported results in animal studies.

- **Pharmacokinetics**

Pantoprazole is rapidly absorbed and the maximal plasma concentration is achieved even after one single oral dose of 20 and 40 mg.

For pantoprazole 20 mg, on an average, at about 2.0 hours - 2.5 hours post-administration the maximum serum concentrations of about 1-1.5 µg/ml are achieved and these values remain constant after multiple administration.

For pantoprazole 40 mg, on average, the maximum serum concentrations are approximately 2-3 µg/ml about 2.5 hours post-administration and these values remain constant after multiple administration.

Volume of distribution is about 0.15 l/kg and clearance is about 0.1 l/h/kg.
Terminal half-life is about 1 h. There were a few reported cases of subjects with delayed elimination. Because of the specific binding of pantoprazole to the proton pumps of the parietal cell the elimination half-life does not correlate with the much longer duration of action (inhibition of acid secretion).

Pharmacokinetics does not vary after single or repeated administration. In the dose range of 10 to 80 mg, the plasma kinetics of pantoprazole are linear after both oral and intravenous administration.

Concomitant intake of food or antacids has no influence on AUC, maximum serum concentrations and thus bioavailability. Only the variability of the lag time will be increased by concomitant food intake.

Pantoprazole's plasma protein binding is about 98%. The substance is almost exclusively metabolised in the liver. The main metabolic pathway is demethylation by CYP2C19 with subsequent sulphate conjugation, other metabolic pathway include oxidation by CYP3A4. Renal elimination represents the major route of excretion (about 80%) for the metabolites of pantoprazole; the rest are excreted in the faeces. The main metabolite in both the plasma and urine is desmethylpantoprazole, which is conjugated with sulphate. The half-life of the main metabolites (about 1.5 hours) is not much longer than that of pantoprazole.

Special population

Pediatric

The AUC, C_{max} and volume of distribution of single doses of pantoprazole 20 mg and 40 mg in children aged 5-16 years have been reported to be similar to adults.

There is no significant association between pantoprazole clearance and age or weight.

Geriatric

A slight increase in AUC and C_{max} in elderly volunteers compared to younger counterparts has been reported to be clinically irrelevant.

Renal Impairment

No dose reduction is required when pantoprazole is administered to patients with impaired kidney function (including dialysis patients). As with healthy subjects, pantoprazole's half-life is short. Only very small amounts of pantoprazole are dialysed. Although the main metabolite has a moderately delayed half-life (2-3 hours), excretion is still rapid and thus accumulation does not occur.

Hepatic Impairment

Although for patients with liver cirrhosis (classes A and B according to *Child*) the half-life time values increase to between 3-6 h for pantoprazole 20 mg and 7-9 hours for pantoprazole 40 mg, the AUC values increase by a factor of 3-5 for pantoprazole 20 mg and 5-7 for pantoprazole 40 mg. The maximum serum concentration increase only slightly by a factor of 1.3 and 1.5 for pantoprazole 20 mg and pantoprazole 40 mg respectively, compared with healthy subjects.

INDICATIONS

PANTOCID GR TABLETS 20 MG are used for:

- Treatment of mild reflux disease and associated symptoms (e.g. heartburn, acid regurgitation, pain on swallowing)
- Long-term management and prevention of relapse in reflux esophagitis
- Prevention of gastroduodenal ulcers induced by non-selective non-steroidal anti-inflammatory drugs (NSAIDs) in patients at risk with a need for continuous NSAID treatment

PANTOCID GR TABLETS 40 MG are used for:

- In combination with 2 appropriate antibiotics for the eradication of *Helicobacter pylori* in patients with peptic ulcers to reduce recurrence of duodenal and gastric ulcers caused by this micro-organism
- Duodenal ulcer
- Gastric ulcer
- Moderate and severe cases of inflammation of the esophagus (reflux esophagitis)
- Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions

DOSAGE AND ADMINISTRATION

Unless otherwise prescribed, the following oral dosing information apply. Follow instructions carefully otherwise **PANTOCID GR TABLETS** tablets may not have the desired effect.

PANTOCID GR TABLETS tablets should not be chewed or crushed, and should be swallowed whole with liquid before a meal.

PANTOCID GR TABLETS 20 MG

Adults and adolescents 12 years of age and above:

Mild reflux disease and associated symptoms (e.g. heartburn, acid regurgitation, pain on swallowing)

The recommended oral dosage is one **PANTOCID GR TABLETS 20 mg** tablet per day.

Long-term management and prevention of relapse in reflux oesophagitis

For long-term management, a maintenance dose of one **PANTOCID GR TABLETS 20 mg** tablet per day is recommended, increasing to one **PANTOCID GR TABLETS 40 mg** tablet per day if a relapse occurs. After healing of the relapse the dosage can be reduced again to 20 mg pantoprazole.

Adults

Prevention of gastroduodenal ulcers induced by non-selective non-steroidal anti-inflammatory drugs (NSAIDs) in patients at risk with a need for continuous NSAID treatment

The recommended oral dosage is one **PANTOCID GR TABLETS 20 mg** tablet per day.

Children below 12 years of age

Pantoprazole is not recommended for use in children below 12 years of age due to limited data in this age group.

Special populations

A daily dose of **PANTOCID GR TABLETS 20 mg** should not be exceeded in patients with severe liver impairment.

No dose adjustment is necessary in elderly patients or in those with impaired renal function.

Type and duration of treatment

Mild reflux disease and associated symptoms (e.g. heartburn, acid regurgitation, pain on swallowing)

Symptom relief is generally accomplished within 2-4 weeks, and a 4-week treatment period is usually required for healing of associated oesophagitis. If this is not sufficient, healing will normally be achieved within a further 4 weeks. When symptom relief has been achieved, reoccurring symptoms can be controlled using an on-demand regimen of 20 mg once daily, when required. A switch to continuous therapy may be considered in case satisfactory symptom control cannot be maintained with on-demand treatment.

Long term management and prevention of relapse in reflux oesophagitis

In long term treatment, a treatment period of 1 year should be exceeded only after careful consideration of the benefit/risk ratio, as drug safety over several years is not sufficiently established.

PANTOCID GR TABLETS 40 MG

Adults and adolescents 12 years of age and above

Treatment of moderate and severe reflux oesophagitis

One tablet of **PANTOCID GR TABLETS 40 mg** tablet per day. In individual cases the dose may be doubled (increase to 2 tablets of **PANTOCID GR TABLETS 40 mg)** especially when there has been no response to other treatment.

Adults

Eradication of H. pylori in combination with two appropriate antibiotics:

In cases of duodenal or gastric ulcer in which infection with *Helicobacter pylori* has been confirmed, the micro-organism should be eradicated by combination treatment. Depending on the resistance pattern, the following combinations are recommended:

- 2 x 1 **PANTOCID GR TABLETS 40 mg** tablet/day + 2 x 1000 mg amoxicillin/ day + 2 x 500 mg clarithromycin/ day
- 2 x 1 **PANTOCID GR TABLETS 40 mg** tablet/day + 2 x 500 mg metronidazole/ day + 2 x 500 mg clarithromycin/ day
- 2 x 1 **PANTOCID GR TABLETS 40 mg** tablet/day + 2 x 1000mg amoxicillin/ day+ 2 x 500 mg metronidazole/ day

Treatment of gastric and duodenal ulcer

One tablet of **PANTOCID GR TABLETS 40 mg** per day. In individual cases the dose may be doubled (increase to 2 tablets of **PANTOCID GR TABLETS 40 mg**) especially when there has been no response to other treatment.

Zollinger-Ellison syndrome and other pathological hypersecretory conditions

For the long-term management of Zollinger-Ellison syndrome and other pathological hypersecretory conditions patients should start their treatment with a daily dose of 80 mg (2 tablets of **PANTOCID GR TABLETS 40 mg**). Thereafter, the dosage can be titrated up or down as needed using measurements of gastric acid secretion to guide. With doses above 80 mg daily, the dose should be divided and given twice daily. A temporary increase of the dosage above 160 mg pantoprazole is possible but should not be applied longer than required for adequate acid control.

Treatment duration in Zollinger-Ellison syndrome and other pathological hypersecretory conditions is not limited and should be adapted according to clinical needs.

Children below 12 years of age:

PANTOCID GR TABLETS 40 MG is not recommended for use in children below 12 years of age due to limited information in this age group.

Type and duration of treatment

Combination therapy for eradication of Helicobacter pylori infection usually lasts 7 days and can be extended to a maximum of 2 weeks. If after this time further treatment with **PANTOCID GR TABLETS 40 mg** is indicated to ensure that the ulcer heals completely, the dose recommendation for gastric and duodenal ulcers must be observed.

In the majority of cases, a duodenal ulcer heals completely within 2 weeks. If a two week treatment period is not sufficient, healing will be achieved in almost all cases within a further 2 weeks. Gastric ulcers and reflux esophagitis usually require a 4 week course of treatment. If this should be inadequate, healing will in most cases be achieved within a further 4 weeks.
Treatment should not exceed 8 weeks as experience with long term use is limited.

Treatment duration in Zollinger Ellison Syndrome and other pathological hypersecretory conditions is not limited and should be adapted according to clinical needs

CONTRAINDICATIONS

PANTOCID GR TABLETS are contraindicated in patients with known hypersensitivity to pantoprazole, any substituted benzimidazole or any component of the formulation.

Pantoprazole like other PPIs should not be co-administered with atazanavir.
Pantoprazole must not be used in combination treatment for eradication of *H. pylori* in patients with moderate to severe hepatic or renal dysfunction since currently no information is available on the efficacy and safety of pantoprazole in combination treatment of these patients.

WARNINGS AND PRECAUTIONS

In patients with severe liver impairment the liver enzymes should be monitored regularly during treatment with pantoprazole, particularly on long-term use. In the case of a rise of the liver enzymes the treatment should be discontinued.

The use of pantoprazole as a preventive of gastroduodenal ulcers induced by non-selective non-steroidal anti-inflammatory drugs (NSAIDs) should be restricted to patients who require continued NSAID treatment and have an increased risk to develop gastrointestinal complications.

Symptomatic response to therapy with pantoprazole does not preclude the presence of gastric malignancy.

Atrophic gastritis has been noted occasionally in gastric corpus biopsies from patients treated long-term with pantoprazole, particularly in patients who were *H. pylori* positive.

The increased risk should be assessed according to individual risk factors, e.g. high age (>65 years), history of gastric or duodenal ulcer or upper gastrointestinal bleeding. Pantoprazole, as all acid-blocking medicines, may reduce the absorption of vitamin B12 (cyanocobalamin) due to hypo- or achlorhydria. This should be considered in patients with reduced body stores or risk factors for reduced vitamin B12 absorption on long-term therapy and in patients with Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions requiring long-term treatment or if respective clinical symptoms are reported.

In long-term treatment, especially when exceeding a treatment period of 1 year, patients should be kept under regular surveillance.

In the presence of any alarm symptom (e. g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis, anaemia or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded, as treatment with pantoprazole may alleviate symptoms and delay diagnosis.

Further investigation is to be considered if symptoms persist despite adequate treatment.

Pantoprazole, like all proton pump inhibitors, might be expected to increase the counts of bacteria normally present in the upper gastrointestinal tract. Treatment with Pantoprazole may lead to a slightly increased risk of gastrointestinal infections caused by bacteria (e.g. *Salmonella*, *Campylobacter*, and *C. difficile*).

PPI therapy may be associated with an increased risk for osteoporosis related fractures of the hip, wrist, or spine. The risk of fracture was reported to increase in patients who received high dose, defined as multiple daily doses , and long term PPI therapy (a year or longer). Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated. Patients at risk for osteoporosis related fractures should be managed according to established treatment guidelines.

It has been reported that long-term exposure of pantoprazole was carcinogenic and caused rare types of gastrointestinal tumors in rodents. The relevance of these findings to tumor development in humans is unknown.

Hypomagnesemia, symptomatic and asymptomatic, has been reported in patients treated with PPIs for at least three months, in most cases after a year of therapy. Serious adverse events include tetany, arrhythmias, and seizures. In most patients, treatment of hypomagnesemia required magnesium replacement and discontinuation of the PPI. For patients expected to be on prolonged treatment or who take PPIs with medications such as digoxin or drugs that may cause hypomagnesemia (e.g., diuretics), health care professionals may consider monitoring magnesium levels prior to initiation of PPI treatment and periodically.

Literature suggests that concomitant use of PPIs with methotrexate (primarily at high dose; see methotrexate prescribing information) may elevate and prolong serum levels of methotrexate and/or its metabolite, possibly leading to methotrexate toxicities. In high-dose methotrexate administration, a temporary withdrawal of the PPI may be considered in some patients.

Effects on ability to drive and use machines

Adverse drug reactions such as dizziness and visual disturbances may occur (see **UNDESIRABLE EFFECTS**). If affected, patients should not drive or operate machines.

DRUG INTERACTIONS

Pantoprazole may reduce the absorption of medicinal products whose bioavailability is pH-dependent (e.g. ketoconazole, other azole antifungals as itraconazole, posaconazole; medicines as erlotinib, ampicillin esters, and iron salts).

It has been reported that co-administration of atazanavir 300 mg/ritonavir 100 mg with omeprazole (40 mg once daily) or atazanavir 400 mg with lansoprazole (60 mg single dose) to healthy volunteers resulted in a substantial reduction in the bioavailability of atazanavir. The absorption of atazanavir is pH dependent. Therefore PPIs, including pantoprazole, should not be co-administered with atazanavir.

Co-administration of atazanavir or nefinavir with PPI is expected to substantially decrease atazanavir or nefinavir plasma concentrations and may result in a loss of therapeutic effect and development of drug resistance. Concomitant use of atazanavir or nelfinavir with proton pump inhibitors is not recommended (see **CONTRAINDICATIONS**).

Pantoprazole is metabolised in the liver via the cytochrome P450 enzyme system. An interaction of pantoprazole with other drugs or compounds, which are metabolised using the same enzyme system, cannot be excluded. However, no clinically significant interactions were reported in specific tests with a number of such drugs/compounds, namely carbamazepine, caffeine, diazepam, diclofenac, digoxin, ethanol, glibenclamide, metoprolol, naproxen, nifedipine, phenytoin, piroxicam, theophylline and an oral contraceptive containing levonorgestrel and ethinyl oestradiol. No interactions have been reported with concomitant administration of pantoprazole with antacids.

Although no pharmacokinetic interaction during concomitant administration of phenprocoumon or warfarin has been reported, a few isolated cases of changes in INR during concomitant treatment have been reported. Therefore, in patients treated with coumarin anticoagulants, monitoring of prothrombin time/INR is recommended after initiation, termination or during irregular use of pantoprazole.

No clinically relevant interactions have been reported with concomitant administration of pantoprazole with respective antibiotics (clarithromycin, metronidazole, amoxicillin).

False positive urine screening tests for tetrahydrocannabinol (THC) in patients receiving PPI have been reported. An alternative confirmatory method should be considered to verify positive results.

Concomitant administration of pantoprazole and clopidogrel had no clinically important effect on exposure to the active metabolite of clopidogrel or clopidogrel induced platelet inhibition. No dose adjustment of clopidogrel is necessary when administered with an approved dose of pantoprazole.

Case reports and published literature suggest that concomitant administration of PPIs and methotrexate (primarily at high dose; see methotrexate prescribing information) may elevate and prolong serum levels of methotrexate and/or its metabolite hydroxymethotrexate. However, no formal drug interaction studies of methotrexate with PPIs have been conducted (see **WARNINGS AND PRECAUTIONS**).

SIDE EFFECTS/ADVERSE REACTIONS

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Frequency	**Very common**	**Common**	**Uncommon**	**Rare**	**Very rare**
System Organ Class					
Blood and lymphatic system				Agranulocytopenia	Thrombocytopenia; Leukopenia Pancytopenia
Nervous system disorders			Headache; Dizziness	Taste disorders	
Eye disorders				Disturbances in vision / blurred vision	

Frequency System Organ Class	Very common	Common	Uncommon	Rare	Very rare
Gastrointestinal Disorders			Diarrhoea; Nausea/ vomiting; Abdominal distension and bloating; Constipation; Dry mouth; Abdominal pain and discomfort		
Skin and sub-cutaneous tissue disorders			Rash/ exanthema /eruption; Pruritus	Urticaria; Angioedema	
Musculoskeletal, connective tissue disorders				Arthralgia; Myalgia	
Metabolism and nutrition disorders				Hyperlipidaemias and lipid increases (triglycerides, cholesterol); Weight changes	
General disorders and administration site conditions			Asthenia, fatigue and malaise	Body temperature increased; Oedema peripheral	
Immune system disorders				Hypersensitivity (incl. anaphylactic reactions and anaphylactic shock)	
Hepatobiliary disorders			Liver enzymes increased (transaminases,	Bilirubin increased	

Frequency System Organ Class	Very common	Common	Uncommon	Rare	Very rare
			γ-GT)		
Psychiatric disorders			Sleep disorders	Depression (and all aggravations)	Disorientation (and all aggravations)
Reproductive system and breast disorders:				Gynaecomastia	

The following additional undesirable effects have been reported:
Hepatobiliary disorders: Hepatocellular injury, jaundice, hepatocellular failure, hepatitis
Psychiatric disorders: Hallucination, Confusion (especially in pre-disposed patients, as well as the aggravation of these symptoms in case of pre-existence), insomnia, somnolence
Renal and urinary disorders: Interstitial nephritis
Skin and subcutaneous tissue disorders: Stevens-Johnson syndrome, Lyell syndrome; Erythema multiforme, photosensitivity, toxic epidermal necrosis (TEN, some fatal)
Body as a whole: Facial oedema, allergic reaction
Nervous: Vertigo
Metabolic/Nutritional: elevated CK (creatine kinase), generalized oedema, hyponatraemia, hypomagnesaemia
Gastrointestinal Disorders: flatulence
Musculoskeletal disorders: rhabdomyolysis, bone fracture
Respiratory: URI

USE IN SPECIAL POPULATION

- **Pregnancy**

No adequate information regarding use of pantoprazole in pregnant women is available. Reproductive toxicity has been reported in animals. The potential risk for humans is unknown. Pantoprazole should not be used during pregnancy unless clearly necessary.

- **Lactation**

It is unknown whether pantoprazole is excreted in human breast milk. Pantoprazole is reported to be excreted in breast milk in animals.

A decision on whether to continue/discontinue breast feeding or to continue/discontinue therapy with pantoprazole should be made taking into account the benefit of breast feeding to the child and the benefit of pantoprazole therapy to the woman.

- **Pediatric**

Safety and effectiveness of pantoprazole in pediatric patients below 12 years of age has not been reported.

- **Geriatric**

Erosive esophagitis healing rates have been reported to be similar in elderly patients (≥ 65 years old) treated with pantoprazole and in patients under the age of 65. The reported incidence of adverse events and laboratory abnormalities in patients aged 65 years and older were similar to those associated with patients younger than 65 years of age.

- **Gender**

Erosive esophagitis healing rates have been reported to be similar in females treated with pantoprazole and in males. Healing was maintained similarly in male and females. The reported incidence of adverse events and laboratory abnormalities in females were similar to those associated with males.

OVERDOSAGE

There are no known symptoms of overdosage in man.
Doses up to 240mg i.v. administered over two minutes were well tolerated. As pantoprazole is extensively protein bound, it is not readily dialyzable. Apart from symptomatic and supportive treatment, no specific therapeutic recommendations can be made.

STORAGE

Store at a temperature below 30^{o}C, protect from moisture.

PACKING

Blister strips of 2 x 7 Tablets, 4 x 7 Tablets, 10 x 10 Tablets.

Manufactured by / Marketing Authorization Holder:
RANBAXY (MALAYSIA) SDN. BHD.
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08000 Sungai Petani, Kedah,
Malaysia.

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